Quadratic Simultaneous Equations

AQA Higher · GCSE Grade 8–9

Name: ___________________ Class: _________ Date: _________

What you need to be able to do

Solve a pair of equations where one is a straight line (linear) and one is a curve (quadratic), using substitution. Most pairs have two answer points — find both x values and the y that goes with each.

Common mistakes to avoid

- There are usually two answers, not one. Find both x values — for example $x = 3$ and $x = -1$.
- Get everything on one side so it equals 0 first. For example: $x^2 = 2x + 3$ becomes $x^2 - 2x - 3 = 0$.
- Watch the minus signs. $(x - 1)^2 = x^2 - 2x + 1$, not $x^2 - 1$.
- Each x has its own y. After you find x, put it back in to work out y.

Method steps

1. Step 1: Make y (or x) the subject of the straight-line equation.
2. Step 2: Substitute that into the quadratic equation.
3. Step 3: Move everything to one side so it equals 0.
4. Step 4: Solve the quadratic — factorise or use the formula.
5. Step 5: Put each value back to find its pair. Write answers as (x, y).

Worked example

Solve: $y = x^2 - 2x + 3$ and $y = x + 3$

Step 1 — Make y the subject of the line: $y = x + 3$ (already done)

Step 2 — Substitute into the quadratic: $x^2 - 2x + 3 = x + 3$

Step 3 — Move everything to one side so it = 0: $x^2 - 3x = 0$

Step 4 — Solve (factorise): $x(x - 3) = 0 \rightarrow x = 0$ or $x = 3$

Step 5 — Put each value back to find its pair: (0, 3) and (3, 6)

Solve the simultaneous equations.

$y = x^2$ and $y = 2x + 3$

How to start

1. First, both equations already give y, so set them equal: $x^2 = 2x + 3$
2. Next, move everything to one side so it equals 0: $x^2 - 2x - 3 = 0$
3. Then factorise and solve to find the two x values.
4. Finally, put each x back into $y = 2x + 3$ to find its y.

x = ______________________

y = ______________________

Remember the method:

1 Make y the subject of the straight line   2 Substitute into the quadratic   3 Rearrange so it = 0

4 Solve (factorise or formula)   5 Find each pair (x, y) and check

[3 marks]

Solve the simultaneous equations.

$y = x^2 + 1$ and $y = x + 3$

How to start

1. Set the two right-hand sides equal, then rearrange to = 0.
2. Factorise to find the two x values, then find each matching y.

x = ______________________

y = ______________________

Remember the method:

1 Make y the subject of the straight line 2 Substitute into the quadratic 3 Rearrange so it = 0
4 Solve (factorise or formula) 5 Find each pair (x, y) and check

Question 3 of 8

[3 marks] ●●○○○

Solve the simultaneous equations.

$y = x^2 - 2x$ and $y = x + 4$

x = ____________________

y = ____________________

Remember the method:

1 Make y the subject of the straight line 2 Substitute into the quadratic 3 Rearrange so it = 0

4 Solve (factorise or formula) 5 Find each pair (x, y) and check

[3 marks]

Solve the simultaneous equations.

$y = x^2 - 5x + 6$ and $y = x - 2$

x = ____________________

y = ____________________

Remember the method:

1 Make y the subject of the straight line 2 Substitute into the quadratic 3 Rearrange so it = 0

4 Solve (factorise or formula) 5 Find each pair (x, y) and check

Question 5 of 8

[4 marks]

Solve the simultaneous equations.

$x^2 + y^2 = 25$ and $y = x - 1$

x = ______________________

y = ______________________

Remember the method:

1 Make y the subject of the straight line   2 Substitute into the quadratic   3 Rearrange so it = 0

4 Solve (factorise or formula)   5 Find each pair (x, y) and check

Question 6 of 8

[4 marks]

Solve the simultaneous equations.

$x^2 + y^2 = 10$ and $x + y = 4$

x = ____________________

y = ____________________

Remember the method:

1 Make y the subject of the straight line   2 Substitute into the quadratic   3 Rearrange so it = 0

4 Solve (factorise or formula)   5 Find each pair (x, y) and check

[5 marks] ●●●●○

Solve the simultaneous equations. Give your answers in surd form.

$x^2 + y^2 = 9$ and $y = x + 1$

x = ______________________

y = ______________________

Remember the method:

1 Make y the subject of the straight line   2 Substitute into the quadratic   3 Rearrange so it = 0

4 Solve (factorise or formula)   5 Find each pair (x, y) and check

[5 marks]

A rectangle has an area of 12 cm^2 and a perimeter of 14 cm.

By forming and solving a pair of simultaneous equations, find the length and width of the rectangle.

(Let the length be x and the width be y.)

x = ____________________

y = ____________________

Remember the method:

1 Make y the subject of the straight line   2 Substitute into the quadratic   3 Rearrange so it = 0

4 Solve (factorise or formula)   5 Find each pair (x, y) and check

Self-reflection

How confident do you feel with this topic now? (tick one)

☐ Not yet ☐ Getting there ☐ Confident ☐ Really confident

Which question did you find the hardest, and why?

One method step I want to remember next time:

One thing I will practise before the next lesson:

Teacher key

Answers

1. Q1: (3, 9) and (-1, 1)
2. Q2: (2, 5) and (-1, 2)
3. Q3: (4, 8) and (-1, 3)
4. Q4: (4, 2) and (2, 0)
5. Q5: (4, 3) and (-3, -4)
6. Q6: (1, 3) and (3, 1)
7. Q7: $x = (-1 + \sqrt{17})/2$, $y = (1 + \sqrt{17})/2$ and $x = (-1 - \sqrt{17})/2$, $y = (1 - \sqrt{17})/2$
8. Q8: length 4 cm, width 3 cm (from $xy = 12$ and $x + y = 7$)